

Certifying and Learning Local Quantum Hamiltonians

arXiv:2603.29809 | TQC 2026

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QNRC Workshop

May 11, 2026



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Motivation

Why learn Hamiltonians?

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This motivates the problem of *Hamiltonian learning*:

Can we learn the Hamiltonian governing an unknown quantum system?

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Suppose a provider claims that their device evolves according to a target Hamiltonian H_0 .

Then the relevant question is simply:

Is the implemented Hamiltonian actually H_0 ?

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This can be much less resource-intensive than full Hamiltonian learning.

Our setting

We study Hamiltonian certification from access to the dynamics

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Throughout the talk, k should be thought of as a constant.

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They are also fundamental in quantum complexity theory.

For instance, the k -local Hamiltonian problem is QMA-complete, and local Hamiltonians are universal for quantum simulation.

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More concretely:

How much total evolution time is necessary and sufficient?

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This achieves *Heisenberg scaling* in the accuracy parameter ε .

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We also show that the protocol can be made *tolerant*.

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The question becomes:

Can we learn or certify this thermal state efficiently?

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This suggests that Gibbs-state learning and certification should be treated as independent tasks.

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This resolves the question of whether efficient certification of Gibbs states is possible in this setting.

Conceptual message

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For dynamics, this leads to an optimal Heisenberg-limited certification protocol.

For thermal states, this lets us avoid the exponential-in- β barrier for Hamiltonian learning.

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We assume without loss of generality that

$$h_{I^{\otimes n}} = \frac{\text{Tr}[H]}{2^n} = 0.$$

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$$\frac{e^{-\beta(H+cI)}}{\text{Tr}[e^{-\beta(H+cI)}]} = \frac{e^{-\beta H}}{\text{Tr}[e^{-\beta H}]}.$$

Thus, assuming traceless Hamiltonians is without loss of generality.

Certification via access to the dynamics

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In the lab, one query to $U_H(t)$ corresponds to letting the system evolve for time t .

Experiments

An experiment can interleave unknown evolution and known controls:

$$\rho \longrightarrow U_H(t_1) \longrightarrow V_1 \longrightarrow U_H(t_2) \longrightarrow V_2 \longrightarrow \cdots \longrightarrow \text{measure}.$$

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For that, please come to Myeongjin's talk tomorrow!

(Joint work with Myeongjin Shin, Changhun Oh, and myself.)

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This is an average-case notion of distance between Hamiltonians.

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So our setting is:

bounded in a worst-case norm, tested in an average-case norm.

Result: Hamiltonian certification

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With high probability, it distinguishes between

$$\|H - H_0\|_F \leq \frac{\varepsilon}{8 \cdot 3^k}$$

and

$$\|H - H_0\|_F \geq \varepsilon.$$

Main result 1 (Formal)

Theorem (Tolerant certification of k -local Hamiltonians)

Let H and H_0 be k -local n -qubit Hamiltonians, where H_0 is known and H can be accessed via its time evolution operator. Let $\varepsilon, \delta > 0$. Let $C_{\text{op}} \geq 1$ be such that $\|H_0\|_{\text{op}}, \|H\|_{\text{op}} \leq C_{\text{op}}$.

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- There is an algorithm that uses $O(9^{2k} \log(1/\delta)/\varepsilon)$ total evolution time to test, with success probability $\geq 1 - \delta$, whether $\|H - H_0\| \leq \varepsilon/(8 \cdot 3^k)$ or $\|H - H_0\|_F \geq \varepsilon$.*

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- Furthermore, the algorithm uses no ancilla qubits, it makes $O(9^{2k} \log(1/\delta))$ experiments, and it makes $O(3^{5k} (C_{\text{op}}/\varepsilon)^{3/2} \log(1/\delta))$ time evolution queries.*

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- Furthermore, the algorithm uses no ancilla qubits, it makes $O(9^{2k} \log(1/\delta))$ experiments, and it makes $O(3^{5k} (C_{\text{op}}/\varepsilon)^{3/2} \log(1/\delta))$ time evolution queries.*
- The algorithm is robust to at least $\Omega(9^{-k})$ SPAM errors, and the classical post-processing time is $O(9^{2k} \log(1/\delta))$.*

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Thus, our result achieves the optimal Heisenberg scaling.

To our knowledge, this gives the first optimal algorithm for testing a property of quantum Hamiltonians.

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For certifying sparse Hamiltonians, a previous approach gives evolution time roughly $\tilde{O}(n^{3k/2}/\varepsilon)$ when specialized to k -local Hamiltonians.

Our algorithm removes this polynomial dependence on n from the total evolution time!

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The goal is to detect whether ΔH is zero or has large Frobenius norm.

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This is where locality of ΔH becomes crucial.

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This lets us perform tests on the difference Hamiltonian ΔH .

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So $I(t)$ depends only on eigenvalue differences of ΔH .

The key spectral quantity

Define the proportion of ε -separated eigenvalue pairs:

$$\Lambda(\Delta H, \varepsilon) = \frac{1}{4^n} \sum_{\substack{r,s: \\ |\lambda_r - \lambda_s| \geq \varepsilon}} 1.$$

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For a uniformly random

$$t \in [0, 2/\varepsilon],$$

we show that

$$I(t) \leq 1 - \frac{\Lambda}{4}$$

with probability at least $1/3$.

From eigenvalue gaps to certification

The previous bound says:

$$\text{many large gaps} \implies I(t) < 1.$$

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So the certification problem reduces to proving that

$$\|\Delta H\|_F \geq \varepsilon$$

implies many large eigenvalue gaps.

Moment calculation

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The fourth moment is

$$\frac{1}{4^n} \sum_{r,s} |F(r, s)|^4 = 2 \|\Delta H\|_4^4 + 6 \|\Delta H\|_2^4.$$

Using hypercontractivity

Because ΔH is k -local, the quantum Bonami lemma gives

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This prevents the eigenvalue-gap distribution from being too concentrated near zero.

Paley–Zygmund step

Applying the Paley–Zygmund inequality gives

$$\Lambda(\Delta H, \|\Delta H\|_F) = \Pr_{r,s} [|\lambda_r - \lambda_s| \geq \|\Delta H\|_F] \geq \frac{1}{3} \cdot 9^{-k}.$$

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This is the key structural fact behind the algorithm.

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If no repetition outputs FAR, output EQUAL.

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Each sampled time satisfies

$$t \leq \frac{2}{\varepsilon}.$$

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For constant k , this is exactly Heisenberg scaling.

Correctness: EQUAL case

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$$\Delta H = 0.$$

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Thus Bell sampling sees non-identity Paulis with noticeable probability, and the algorithm outputs FAR with high probability.

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Our certification algorithm is efficient beyond total evolution time.

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- the number of experiments is independent of the operator norm bound;
- the total evolution time is independent of the operator norm bound;
- the algorithm is robust to constant SPAM error;
- Bell sampling can be replaced by identity-probability estimation;
- hence, the algorithm can be implemented without quantum memory.

Learning thermal states

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A common approach is:

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But Hamiltonian learning from Gibbs states can require sample complexity exponential in β .

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Instead, we learn the Gibbs state directly!

This allows us to obtain sample complexity polynomial in all relevant parameters.

The price is that the learning algorithm is not time-efficient, due to a covering-net search.

Result: Gibbs-state learning

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Then there is an algorithm that ε -learns $\rho_H(\beta)$ in trace norm using

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The guarantee holds with high probability.

Main result 2 (Formal)

Theorem (Learning Gibbs states)

Let ρ be the Gibbs state at inverse temperature β of an n -qubit and k -local Hamiltonian H with $|h_P| \leq 1$ for every P . Let $\delta, \varepsilon \in (0, 1)$.

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Let ρ be the Gibbs state at inverse temperature β of an n -qubit and k -local Hamiltonian H with $|h_P| \leq 1$ for every P . Let $\delta, \varepsilon \in (0, 1)$.

Then, from $O(3^k n^{2k} k \log(n/\delta) (\max\{\beta, 1\})^2 / \varepsilon^4)$ single copies of ρ , our algorithm obtains $\rho' \in \mathcal{S}_{\varepsilon', n, k, \beta}$ such that $\|\rho' - \rho\|_{\text{tr}} \leq \varepsilon$ with probability $\geq 1 - \delta$.

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The classical post-processing time of the protocol is $(n^k \max\{\beta, 1\} / \varepsilon)^{O(n^k)}$.

Why is this useful?

When

$$\beta = \text{poly}(n),$$

this gives Gibbs-state tomography with polynomial sample complexity.

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Thus, for local Gibbs states, we obtain an exponential sample-complexity improvement over general tomography.

The only structural assumption is locality of the Hamiltonian.

Gibbs-state continuity

The proof begins with the inequality

$$\|\rho_H(\beta) - \rho_{H'}(\beta)\|_{\text{tr}} \leq \sqrt{2\beta \text{Tr}[(\rho_H(\beta) - \rho_{H'}(\beta))(H' - H)]}.$$

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For k -local Hamiltonians, this implies

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So coefficient-wise approximation of the Hamiltonian gives trace-norm approximation of the Gibbs state.

Covering net

Choose a grid spacing

$$\eta \asymp \frac{\varepsilon}{\beta n^k}.$$

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Then

$$\mathcal{S}_\eta = \{\rho_H(\beta) : H \in \mathcal{H}_\eta\}$$

is an ε -covering net for local Gibbs states.

Using classical shadows

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$$H - H'$$

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Let $\Delta_{H,H'}$ denote these estimates.

These estimates allow us to compare the unknown state with every candidate Gibbs state in the net.

Selecting a hypothesis

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Formally, choose

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The analysis shows that, even with estimation error and net discretization error,

$$\|\rho - \rho'\|_{\text{tr}} \leq \varepsilon$$

with high probability.

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Therefore, the Gibbs-state learning algorithm is sample-efficient, but not time-efficient.

This motivates the certification problem, where a target state is known.

Certifying thermal states

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Here the goal is both sample efficiency and time efficiency.

Result: Gibbs-state certification

Let $\rho_H(\beta)$ and $\rho_{H_0}(\beta)$ be Gibbs states of local Hamiltonians H and H_0 .

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There is an algorithm that decides whether

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or

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single copies.

Main result 3 (Formal)

Theorem (Certifying Gibbs states)

Let ρ and ρ_0 be the Gibbs states at inverse temperature β of n -qubit and k -local Hamiltonians H and H_0 with $|h_P|, |(h_0)_P| \leq 1$ for every P , respectively. Assume that H_0 is known. Let $\delta, \varepsilon \in (0, 1)$.

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Then, our algorithm decides, with success probability $\geq 1 - \delta$, whether $\|\rho - \rho_0\|_{\text{tr}} \leq \varepsilon^2 / (400\beta n^k)$ or $\|\rho - \rho_0\|_{\text{tr}} \geq 2\varepsilon$ with $O\left(\beta^2 n^{2k} 3^k k \log(n/\delta) / \varepsilon^4\right)$ single copies of ρ and ρ_0 .

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Moreover, the protocol only requires Pauli measurements, and a classical post-processing time of order $O\left(\beta^2 n^{3k} 3^k k \log(n/\delta) / \varepsilon^4\right)$. The same conclusion holds if ρ and ρ_0 are both unknown and we are given copy access to both.

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So the algorithm only needs to test whether the unknown state is close to this target.

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As a result, the certification algorithm is both sample-efficient and time-efficient.

Speedup over general certification

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This resolves the question of whether efficient certification is possible for local Gibbs states.

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samples for learning local Gibbs states.

3. *Gibbs-state certification:* same sample complexity, but also time-efficient.

Discussion and Open Problems

What we have shown

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First, we considered certification from time-evolution access.

We gave an algorithm for local Hamiltonian certification with optimal total evolution time

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To our knowledge, this gives the first optimal bound for any Hamiltonian property testing task in the time-evolution access model.

From Hamiltonians to Gibbs states

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For Gibbs-state learning, we obtained a sample-efficient algorithm in all relevant parameters.

For Gibbs-state certification, we obtained an algorithm that is both sample-efficient and time-efficient.

This overcomes the known exponential-in- β barrier that appears when one first tries to learn the Hamiltonian from Gibbs-state access.

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They are not merely consequences of Hamiltonian learning.

Sometimes, learning the state is easier than learning the Hamiltonian.

Similarly, Hamiltonian certification can be much easier than full Hamiltonian learning.

Open problem I: beyond locality

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For $O(1)$ -local Hamiltonians, we achieve optimal total evolution time

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even without inverse or controlled time evolution.

A fundamental open question is:

*Can one achieve $O(1/\varepsilon)$ certification
without any locality assumption?*

Is locality fundamental?

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Resolving this would clarify whether locality is merely a technical condition in our analysis, or whether it is fundamentally needed for evolution-time-optimal certification.

Open problem II: time-efficient Gibbs-state learning

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This leads to the natural question:

Can Gibbs states of k -local Hamiltonians be learned both sample-efficiently and time-efficiently?

Why this question is natural

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It is natural to ask whether the same evolution is possible for Gibbs-state learning itself.

Open problem III: optimal Gibbs-state certification

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However, we do not know whether the sample complexity is optimal.

Even in the classical setting, the precise complexity of testing local Gibbs distributions appears to be open.

Thus, we ask:

What is the optimal sample complexity of certifying local Gibbs states?

Open problem IV: certification from local probes

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But in many experimental settings, one may only observe a small part of the system.

Recent progress in Hamiltonian learning suggests that even single-site or constant-size probes can sometimes yield meaningful information.

This motivates a certification analogue.

Local probes: main questions

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For example, a constant-size probe may or may not suffice to achieve optimal

$$\Theta(1/\varepsilon)$$

evolution time.

Why local probes matter

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Why local probes matter

Understanding certification from local probes would bring the theory closer to realistic experiments.

In practice, one often has limited control and limited measurement access.

The key question is whether local observations can still detect the spectral features distinguishing

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This remains widely open.

Open problem V: optimal sparsity testing

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One particularly interesting setting is sparsity testing under an additional promise of locality.

Open problem VI: beyond Hamiltonians

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Examples include:

- noisy quantum channels;
- Pauli channels induced by noise processes;
- Lindbladian evolutions for open quantum systems.

A natural direction is to extend Hamiltonian certification techniques to these settings.

Broader direction

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Such a theory should handle:

- locality and sparsity;
- finite-temperature states;
- limited control and local probes;
- noise and open-system dynamics.

This would help bridge the gap between theoretically optimal protocols and experimentally available access models.

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For Gibbs states, it leads to efficient algorithms beyond what Hamiltonian learning alone would imply.

The remaining challenge is to understand how far this separation extends.

Thank you!

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